19 Prove that the sum of the squares of any three consecutive odd numbers is always 11 more than a multiple of 12

Let the three odd numbers be $2n+1$, $2n+3$, $2n+5$,

where n is an integer.

$(2n+1)^2 = 4n^2+4n+1$

1 mark
Expand odd squares

$(2n+3)^2 = 4n^2+12n+9$

$(2n+5)^2 = 4n^2+20n+25$

Sum $= 12n^2 + 36n + 35$

1 mark
Collect all terms

$= 12(n^2 + 3n + 2) + 11.$

Since n^2+3n+2 is an integer, this is 11 more than a multiple of 12.

1 mark
Factor 12; integer reason

(Total for Question 19 is 3 marks)